Problem Set 2

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Problem 2.1 (Sample size). We use the data set `sepe.dat` from Problem Set 1.

a) We plan a new sample survey to estimate total environmental protection expenditures. The survey will use simple random sampling (SRS), and the total will be estimated by $\widehat{t}_y = N\bar{y}$, where $\bar{y}$ is the sample mean.

For sample size planning, assume that future expenditures have approximately the same coefficient of variation as the expenditures observed in `sepe.dat`. Determine the sample size so that the relative error exceeds $\rho = 10\%$ with probability at most $\alpha = 5\%$.

For the population mean, the requirement is

$$\mathbb{P}\left(\left|\frac{\bar{y} - \bar{y}_U}{\bar{y}_U}\right| > \rho\right) \leq \alpha.$$

Rewrite this condition for $\widehat{t}_y$. What is the minimum required sample size?

b) Use the minimum sample size from part (a) to estimate total environmental protection expenditures and report a 95% confidence interval. Cochran proposed the rule $n > 25G_1^2$, where G_1 is Fisher's coefficient of skewness, i.e., the third standardized moment.[1] Sugden et al. proposed the modified rule $n > 28 + 25G_1^2$.[2]

Does the sample size satisfy Cochran's rule? Does it satisfy the rule proposed by Sugden et al.? What do these rules suggest about the adequacy of the 95% confidence interval?

> **Hint.** You can use the following function to compute the finite-population moment coefficient of skewness.

```
skewness_pop <- function(x)
{
    z <- x - mean(x)
    mean(z^3) / mean(z^2)^(3 / 2)
}
```

c) Under the same assumptions as in part a), determine the sample size so that the coefficient of variation of the estimator

$$\mathsf{CV}(\widehat{t}_y) = \frac{\mathsf{SE}(\widehat{t}_y)}{\mathbb{E}(\widehat{t}_y)}$$

is at most 5 %, where $\widehat{t}_y = N\bar{y}$.

d) How large must the sample be to estimate the population proportion P_{31} of establishments in industry class 31 (indicator W_{31}) with a standard error of at most 5 percentage points? Use

$$\widehat{P}_{31} = \frac{1}{n}\sum_{i \in s} \mathbb{1}\{i \in W_{31}\},$$

[1] Cochran, W. G. (1977). Sampling Techniques. 3rd ed., New York: John Wiley & Sons, p. 42.
[2] Sugden, R. A., Smith, T. M. F., and Jones, R. P. (2000). Cochran's Rule for Simple Random Sampling. Journal of the Royal Statistical Society Series B, 62, pp. 787–793.

where

$$\mathbb{1}\{i \in W_{31}\} = \begin{cases} 1 & \text{if } i \text{ belongs to industry class 31,} \\ 0 & \text{otherwise.} \end{cases}$$

Problem 2.2 (Simple random sampling). We use the data set `sepe.dat` from Problem Set 1.

a) Draw a simple random sample without replacement (SRS) of size $n = 200$ from the `sepe.dat` population, which is of size $N = 800$. Estimate

   i) the mean sepeironmental protection expenditure per establishment, $\bar{y}_U$;

   ii) total expenditure, t_y;

   iii) mean expenditure per employee among establishments with at least one employee,

$$\bar{q}_{U+} = \frac{1}{N_+} \sum_{i \in U_+} \frac{y_i}{x_i}, \qquad U_+ = \{i \in U : x_i > 0\}.$$

   Here, y_i denotes expenditure and x_i the number of full-time employees of establishment i. Also estimate the standard error of each estimator.

   Compare the estimated standard errors with the corresponding population-based standard errors. For the sample mean, also compare the estimated variance with the Monte Carlo (MC) variance from Problem 1.3a).

   Hint. Write down the three estimators before doing the calculations in R. To reproduce the solution, use `set.seed(11)` before drawing the sample. A sample of indices can then be obtained with

```
N <- NROW(sepe)
in_sample <- sample(N, 200)
sepe_sample <- sepe[in_sample, ]
```

b) Repeat the three estimates from part a) using the R package `survey` and the sample `sepe_sample`.

   Hint. Attach the `survey` package to the search path, add the population size N to the sample data, and define the survey design object `dn_sepe`:

```
library("survey")
sepe_sample$N <- N
dn_sepe <- svydesign(ids = ~1, fpc = ~N, data = sepe_sample)
```

   The argument `ids = ~1` specifies a design without clustering. Supplying `fpc = ~N` incorporates sampling without replacement from a population of size N.

   New variables can be added to a design object with `update()`. A design object can be restricted to a subpopulation using `subset()`.

   Use `svymean()` to estimate population means and `svytotal()` to estimate population totals. Their basic syntax is

```
svymean(~variable, design)
svytotal(~variable, design)
```

The first argument is an R formula specifying the variable to be analyzed. The tilde `~` is part of the formula syntax; for example, `~expend` specifies the variable `expend`. The second argument is the survey design object containing the sample data and the information about the sampling design.

The functions return objects containing both the estimates and information about their sampling variability. The corresponding methods `coef()`, `SE()`, and `cv()` extract the point estimates, standard errors, and coefficients of variation, respectively.

c) Using the design from part b), construct a 95 % Wald and a logit-based confidence interval for the population proportion P_{31} of establishments in industry class 31, identified by `code = 1`. Use `svyciprop()`.

Hint. A normal-based Wald interval on the probability scale is obtained with

```
svyciprop(~code, dn_sepe, method = "mean", level = 0.95, df = Inf)
```

Setting `df = Inf` specifies that the confidence interval should use the standard normal distribution rather than a Student t-distribution based on the design degrees of freedom.

Using `method = "logit"` instead gives a logit-based confidence interval. The logit interval respects the natural bounds of a proportion and can better accommodate asymmetry when the estimated proportion is close to 0 or 1.

Problem 2.3 (Stratified simple random sampling). Use the file `sepe.dat` again. Define the stratum indicator `strat = 1` for establishments with `empl` ≤ 250 and `strat = 2` otherwise. Also define `Nh`, which gives the population size of the corresponding stratum for each establishment. Add both variables to the data set.
Draw a stratified simple random sample without replacement, with $n_1 = 100$ establishments from the first stratum and $n_2 = 50$ from the second stratum. Estimate the population mean of environmental expenditures and its standard error.
Finally, compare the estimated standard error with that from a simple random sample of $n = 150$ establishments. For the simple random sample, compute the standard error both with and without the finite population correction.

Hint. Load the data `sepe.dat` data and define `strat` and `Nh` as described above. Attach the `survey` package to the search path and draw the stratified sample:

```
library("survey")
set.seed(11)
sepe_sample <- sepe[stratsample(sepe$strat, c("1" = 100, "2" = 50)), ]
```

The function `stratsample()` is provided by the `survey` package. To reproduce the solution, use `set.seed(11)` before drawing the sample. Next, define the survey design object by

```
dn_sepe <- svydesign(ids = ~1, strata = ~strat, fpc = ~Nh,
                     data = sepe_sample)
```

Then use `svymean()` to estimate the population mean and `SE()` to extract its estimated standard error.